

Practical-4

JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Theory

1.How are DOM elements selected and modified?

Answer: DOM elements are selected using methods like getElementById().They are modified using properties like textContent, style, and classList.

2. Are event listeners attached correctly?

Answer: Yes, event listeners are attached using addEventListener() for actions like button clicks, form submission, and menu toggling.

3.Is localStorage used for remembering theme choice?

Answer: Yes, localStorage is used to save the user's selected theme so that the theme remains the same after refreshing or reopening the website.

4. Does interactivity improve usability without breaking accessibility?

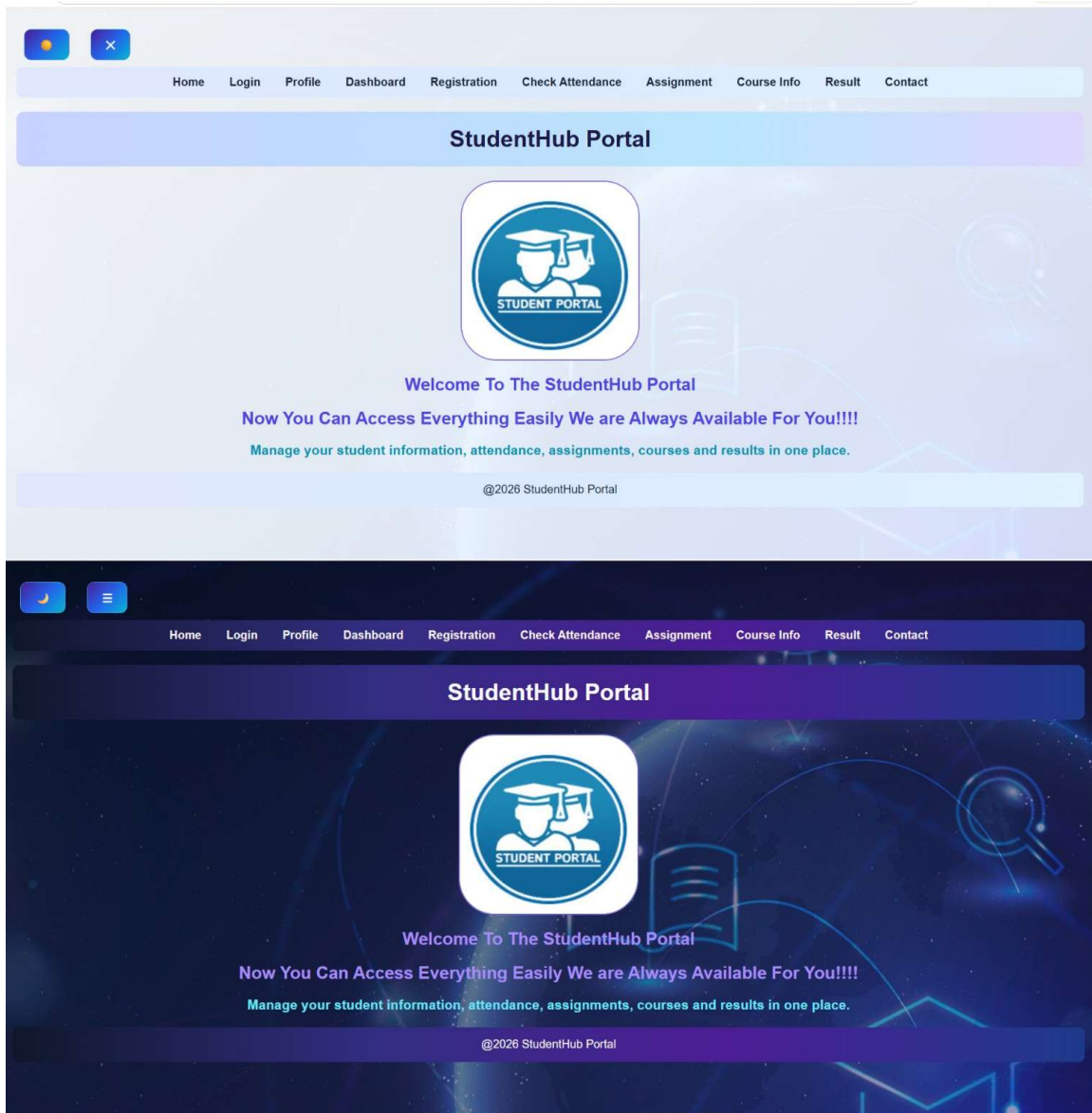
Answer: Yes, features like form validation, theme switching, notifications, and menus improve usability while maintaining accessibility through proper buttons, labels, keyboard support, and readable colors.

A. Result:-

CSS Code:-

<https://github.com/Niket-Borad/StudentHub-Portal/blob/main/Practical%204/script.js>

Evidence of Work (Screenshots)



Student Login

Please enter both username and password.

Login Portal

Username:

Password:

[Submit](#)

[Back to Home](#)

@2026 StudentHub Portal

Student Registration

Please fill in all required fields.

Student Information

Full Name:

Student ID:

Date of Birth: 

Gender

☐ Male ☐ Female

Contact Information

Mobile Number:

Email:

Student Registration

Form cleared successfully!

Student Information

Full Name:

Student ID:

Date of Birth: 

Gender

☐ Male ☐ Female

Contact Information

Mobile Number:

Email:

Contact Us

Please enter both name and message.

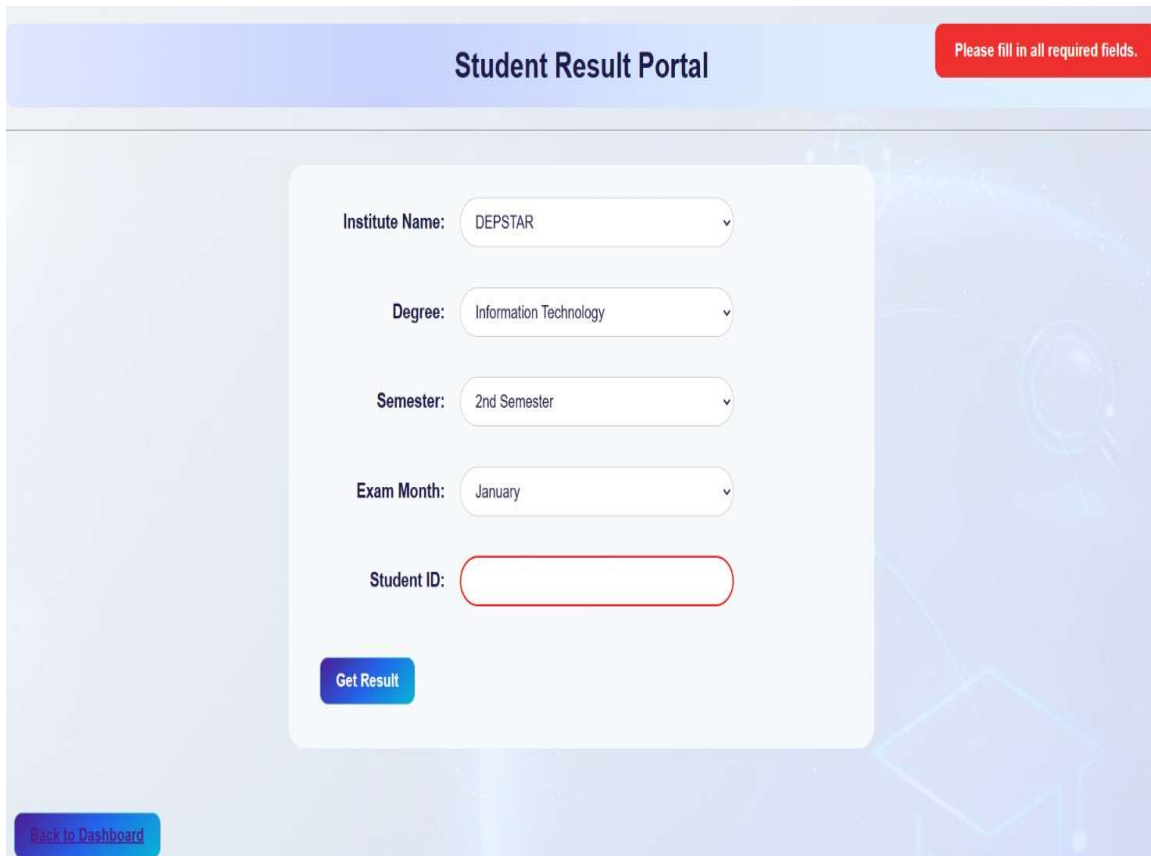
Email: studenthub@gmail.com
Phone: 9876543210
Address: StudentHub College

Send Message

Your Name:

Your Message:

[Back to Dashboard](#) [Back to Home](#)



The screenshot shows a web form titled "Student Result Portal". At the top right, a red notification box says "Please fill in all required fields." The form contains five dropdown menus: "Institute Name" (DEPSTAR), "Degree" (Information Technology), "Semester" (2nd Semester), and "Exam Month" (January). Below these is a text input field for "Student ID", which is currently empty and has a red border. A blue "Get Result" button is positioned below the Student ID field. In the bottom left corner, there is a blue button labeled "Back to Dashboard". The background of the page features a faint, stylized graphic of a graduation cap and some abstract lines.

B. GitHub Repository Link:

<https://github.com/Niket-Borad/StudentHub-Portal>

C. Challenges Faced:

- Understanding how to select and modify HTML elements using the DOM.
- Handling different events like button clicks, form submission, and menu toggling.
- Implementing the light/dark theme switcher using JavaScript and `localStorage`.
- Making the hamburger menu, FAQ, popup, slider, and notification work correctly.
- Maintaining proper styling and accessibility while adding interactivity.
- Fixing errors and testing the JavaScript code using the browser console.

D. Learning Outcomes:

Learned how to select and modify HTML elements using JavaScript DOM methods.

Learned how to use event listeners to make web pages interactive.

Learned how to create dynamic features such as FAQ, popup, slider, notification, and hamburger menu.

Learned how to use localStorage to remember the user's theme preference.

Learned how JavaScript, HTML, and CSS work together to create an interactive website.

Improved debugging and testing skills using browser developer tools

Faculty Remarks: